Confidential Claim Infrastructure for Receivable Finance

A privacy-preserving multi-lender protocol for confidential partial financing and cross-entity fraud containment.

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Executive Summary

Receivable financing allows small and medium enterprises (SMEs) to turn unpaid invoices into immediate working capital. However, the domestic financing ecosystem currently suffers from a critical structural vulnerability: lenders operate in strict information silos. A bank, a non-bank financial institution (NBFI), or a fintech company can maintain flawless internal records but remain entirely blind to whether a specific receivable has already been pledged to a competitor. This visibility gap enables duplicate financing, a recognized global trade-finance fraud pattern that introduces severe risk and silently inflates the cost of credit for all SMEs.

Bangladesh has already legislated a solution to this exact problem. The Secured Transactions (Movable Property) Act of 2023 established the legal mandate for a national collateral registry for movable assets, including receivables. Yet, years later, the market still lacks a working implementation. The barrier is not legal, but architectural: competing financial institutions are fundamentally reluctant to share sensitive client, pricing, and position data with a centralized registry or each other.

FinClaim resolves this coordination failure. It is a permissioned, regulator-supervised receivable claim registry that allows competing lenders to verify asset safety without exposing proprietary data. Deliberately avoiding the naive approach of putting sensitive commercial documents on a public database, FinClaim keeps all actual invoices and business terms strictly encrypted off-chain. The shared distributed ledger acts exclusively as a state machine, recording only the minimum cryptographic evidence necessary to prove identity, state, authorization, and available claim capacity.

By integrating four core mechanisms - a legally-aware claim object, confidential partial-financing checks, a linked-entity fraud containment engine, and an independently verifiable public audit anchor - FinClaim allows any authorized lender to instantly answer one crucial question: *has this receivable already been claimed, and how much financing capacity is still safely available?* The result is a resilient digital infrastructure that protects institutional confidentiality, aligns with the government's digital economy priorities, and safely unlocks vital liquidity for the SME sector.

Introduction

Trade finance runs on a simple promise: a business ships goods or delivers a service, issues an invoice, and gets paid weeks or months later. Factoring allows an SME to turn that promise into cash today instead of waiting 30, 60, or 90 days. This mechanism is critical because a business can be fully profitable on paper and still run out of working capital while it waits to get paid.

However, the vulnerability in this system isn't the invoice itself. It is the fact that an invoice is merely a piece of paper, a PDF, or an isolated database entry. A bank, an NBFI, and a fintech can each keep flawless internal records and still have no way of knowing if the same receivable was already pledged elsewhere, because each institution's view of the world stops at its own ledger.

This is not a hypothetical flaw: duplicate financing is a recognized trade-finance fraud pattern globally, and industry bodies have repeatedly called for shared registries to catch it. Bangladesh already has the foundational pieces to solve this - established receivable-financing products, modern digital banking infrastructure, and real experience running blockchain-based letters of credit. What is missing is the architectural layer that ties them together to establish absolute trust between competing lenders.

Problem Statement

FinClaim addresses a structural coordination problem in domestic receivable financing: lenders operate in separate information silos while the underlying receivable may be presented to more than one institution.

1. The same receivable can be presented to multiple lenders before any of them can see the other claim.
2. A genuine invoice can still become risky if its financing status is hidden across institutions.
3. Simple document hashing does not solve partial financing, because lenders also need to know whether the total claimed amount exceeds the receivable value.
4. Sharing full invoice values and lender positions creates a confidentiality problem between competitors.
5. Fraud can spread across linked companies or repeated transactions, so checking only one invoice at a time may miss a wider pattern.
6. Bangladesh has experienced major banking fraud involving falsified documents and multiple related entities; these cases are not proof of domestic factoring double-pledging, but they show why cross-institution verification and auditability matter.
7. Bangladesh has already legislated a solution to this exact problem. The Secured Transactions (Movable Property) Act, passed in 2023, established a registration authority for movable-asset collateral, including receivables. Three years on, practitioners were still publicly calling for a working national collateral registry as recently as weeks before this paper was written. The gap FinClaim addresses is not the absence of a legal mandate - it is the absence of a working implementation of one.

The FinClaim Solution

FinClaim is a permissioned, regulator-supervised receivable claim registry. It records the financing state of a receivable, not the full commercial document. The actual invoice and business terms remain encrypted off-chain; the ledger stores only the minimum information needed to prove identity, state, authorization, and claim capacity.

Before a receivable becomes financeable, the seller submits it, the buyer confirms that the debt is real, and available identity/tax data can be checked through an oracle gateway. Lenders then create a pledge or assignment claim. The protocol atomically prevents claims from exceeding the receivable's allowed capacity, while a privacy layer is designed to return an eligibility result without exposing competitors' exact positions.

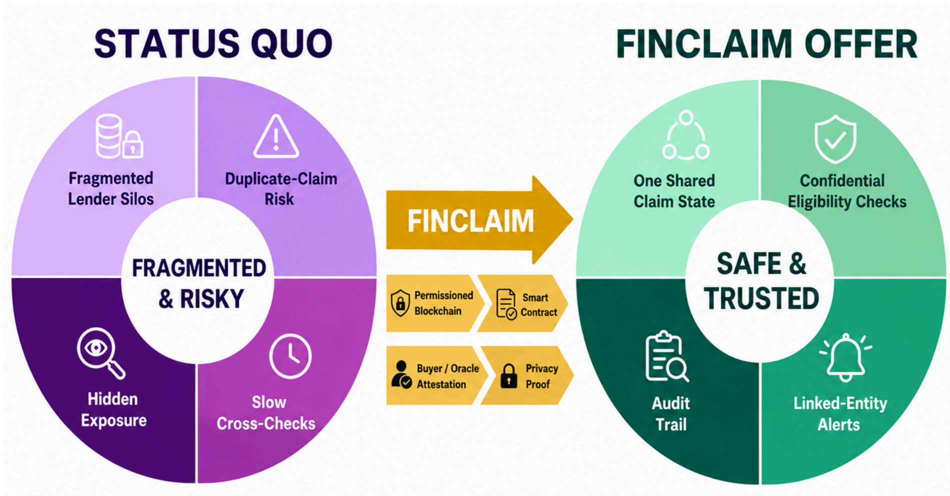
The core contribution is therefore not **"put an invoice on blockchain."** FinClaim combines four mechanisms around a Bangladesh-specific financial workflow:

(1) a legally-aware receivable claim object that distinguishes pledge from assignment,

(2) confidential partial-financing checks,

(3) linked-entity fraud containment, and

(4) a supervisory consortium model with an independently verifiable public audit anchor.



Impact

A safe receivable registry can improve the confidence with which institutions finance SMEs. The direct beneficiary is not "blockchain users"; it is a business that can turn a verified future payment into working capital more safely, and a lender that can verify the asset without asking competitors to reveal client-sensitive details.

FinClaim's expected impact is concentrated in four areas: safer SME liquidity, lower duplicate-financing risk, better cross-institution auditability, and a reusable digital claim infrastructure that can later support other assets with the same "one right, multiple potential claimants" problem.

Alignment with Bangladesh Government Priorities (2026):

The current BNP-led government entered office in February 2026. Its **technology platform** and **FY2026-27 policy** direction emphasize **a larger digital economy**, **ICT-led employment, cash-light digital services, financial inclusion**, and **reform of the banking system**. FinClaim is not a government project and does not claim official endorsement; the alignment below shows where the proposed infrastructure supports stated public-policy goals.

Current Government Technology and Economic Priorities:

Digital economy as a growth engine:

The 2026 BNP technology roadmap set a target for ICT to contribute roughly 5-10% of GDP by 2036 and proposed one million ICT-related jobs, including roles in cybersecurity, AI, business-process outsourcing and other digital services. FinClaim fits this direction as a locally designed financial-infrastructure product rather than a consumer-facing crypto application.

Cash-light and interoperable services:

The same roadmap proposed a national e-wallet and stronger integration with digital payment platforms. FinClaim does not replace payment rails; it complements them by digitizing the trust and claim layer that can sit before financing and settlement.

Financial inclusion and banking confidence:

The FY2026-27 budget speech highlights financial inclusion and growth of the digital economy, while the government has also emphasized banking-sector discipline and accountability. FinClaim aligns by giving lenders a shared fraud-prevention and audit mechanism for asset-backed SME credit.

Local digital capability and secure data:

The technology roadmap has also stressed local cloud/data infrastructure and cybersecurity. FinClaim therefore uses a permissioned architecture, keeps raw invoices off-chain, and treats key management, access control, and auditability as first-class requirements.

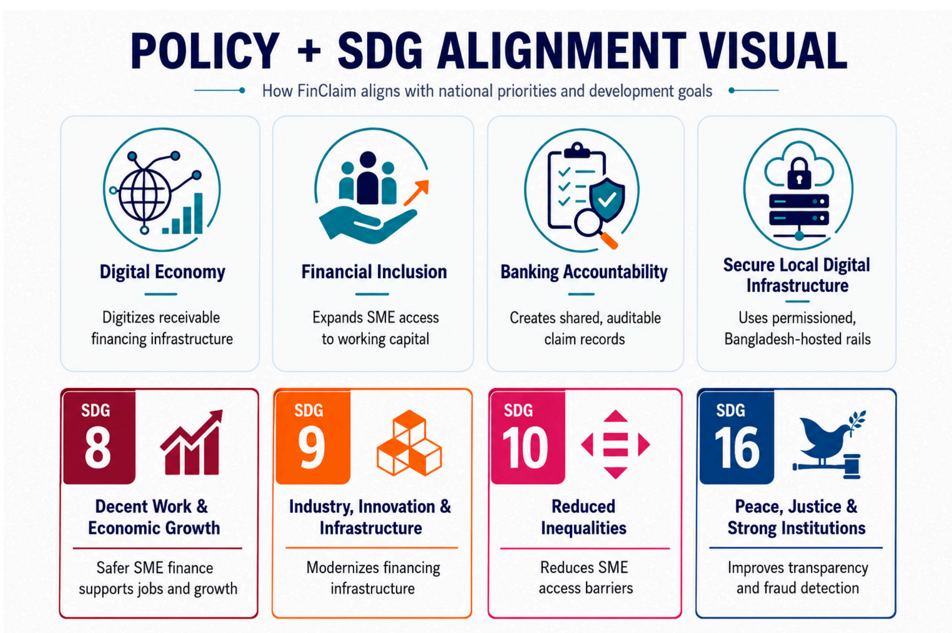
SDG Goals:

SDG 8 (Decent Work and Economic Growth): More reliable working-capital financing can help viable SMEs bridge payment delays, maintain operations and support employment.

SDG 9 (Industry, Innovation and Infrastructure): FinClaim proposes shared digital financial infrastructure that connects institutions without forcing them to surrender control of sensitive business data.

SDG 10 (Reduced Inequalities): Lower verification friction can make smaller firms and less-connected borrowers easier to assess, although inclusion still depends on lender policy and pricing.

SDG 16 (Peace, Justice and Strong Institutions): Immutable event logs, supervisory access and external audit anchoring support transparent institutional processes and stronger fraud investigation evidence.



Why blockchain?

A conventional central database could solve the basic duplicate-claim problem if every lender trusted one operator and regulation forced universal participation. FinClaim does not claim that blockchain is the only possible technology. The blockchain case is narrower: several competing institutions need one state machine, no single lender should be able to rewrite shared history, and the rules for creating, claiming, releasing and settling a receivable should execute consistently for every participant.

This also reframes what FinClaim is asking regulators for. It is not a request for new regulatory attention to a novel idea - it is a proposal to interoperate with, and help fulfil, an existing legal mandate under the 2023 Act. A consortium model succeeds where the Act's single centrally-administered registry has stalled because no individual bank or NBFI has a competitive incentive to feed a shared system accurate, real-time data for the benefit of the sector as a whole. A permissioned ledger with cryptographic accountability per participant removes the excuse, not the requirement, to cooperate - which is the correct scope for what this technology can actually fix.

Opportunities

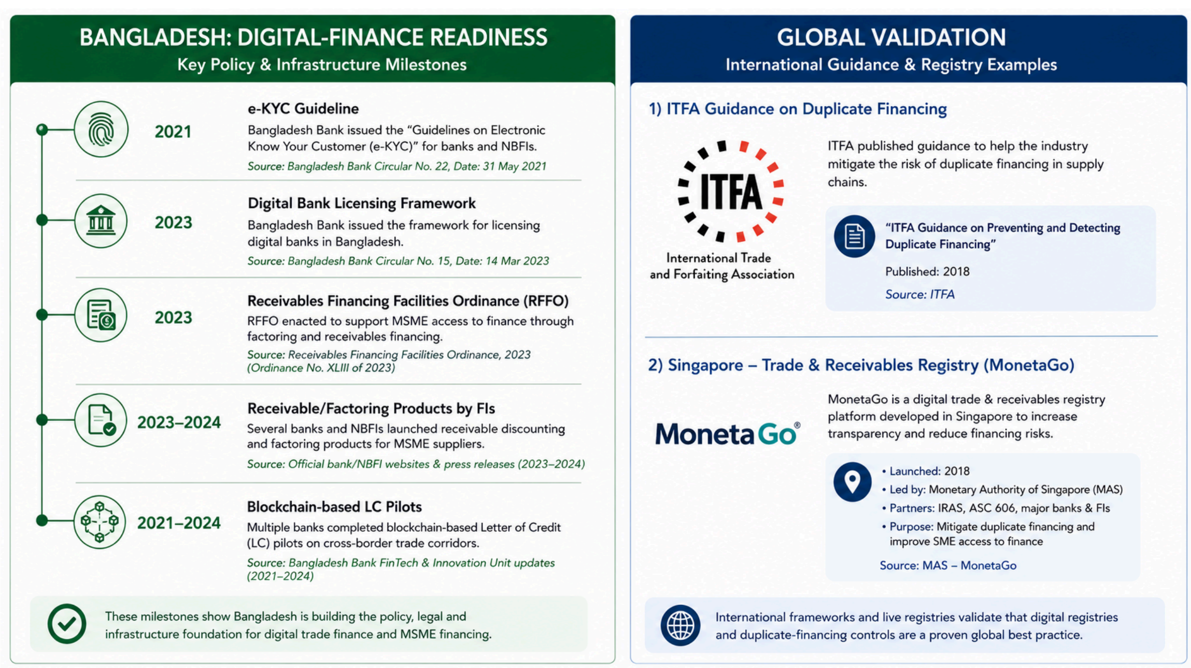
The opportunity is not “blockchain adoption” by itself. It is the gap between growing digital finance and the lack of a common asset-level claim registry.

National Opportunity

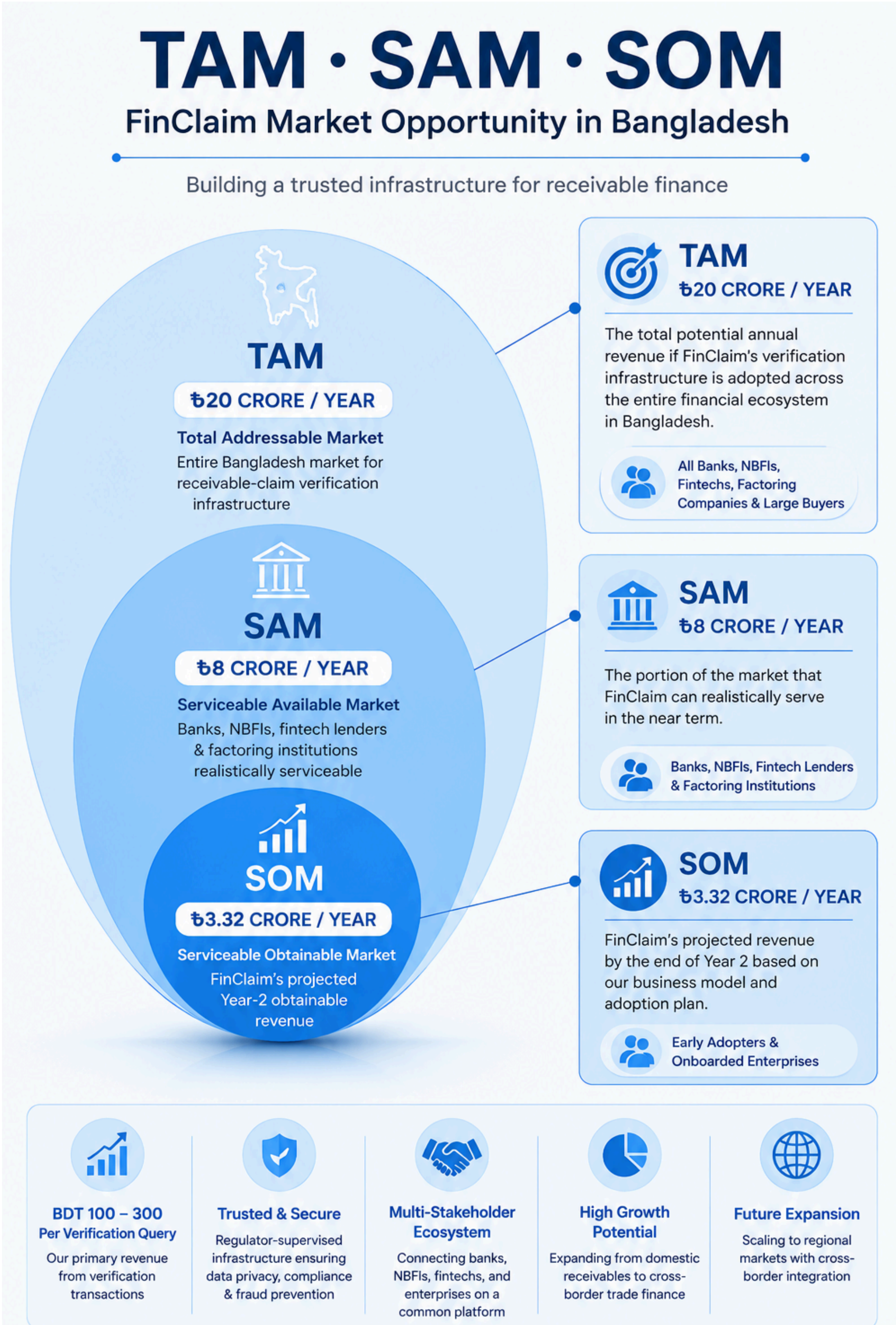
Bangladesh Bank already maintains modern payment infrastructure, e-KYC guidance, digital-bank guidelines, cybersecurity rules and a Regulatory FinTech Facilitation Office (RFFO) that accepts innovative ideas for pilot projects. Banks and NBFIs already offer receivable/factoring products, while Bangladesh has also used blockchain for trade-finance LC workflows. This creates a realistic institutional environment in which a receivable-registry pilot can be tested without pretending the market must be created from zero.

International Opportunity

Duplicate financing is an acknowledged international trade-finance risk. ITFA’s fraud-prevention work recommends wider use of third-party registries that can detect duplicate invoices, and **MonetaGo’s** Secure Financing platform demonstrates that shared fraud-prevention registries can operate across multiple financial institutions. These examples validate the problem, while also proving FinClaim is not globally first. Its differentiation is the Bangladesh domestic use case plus privacy-preserving partial claims, pledge-versus-assignment logic, supervisory design and linked-entity containment.



Market Opportunity in Bangladesh:



Key Features

FinClaim is built around the receivable lifecycle rather than around document storage. The next page summarizes the core features and how they work together.

Receivable Claim Object: One digital record represents each verified receivable. It's status, claim type, and cryptographic proof while the actual invoice stays off-chain and encrypted.

Pledge vs. Assignment Logic: FinClaim treats these as legally different events. A pledge means the seller still owns the receivable but has used it as collateral. An assignment means ownership of the right to collect has been transferred. The system tracks which one applies instead of treating every transaction the same way.

Confidential Partial-Financing Proof: Before approving a new claim, the system checks "do existing claims plus this new one stay within the receivable's value?" and returns a simple TRUE/FALSE without revealing exact amounts to other lenders. The prototype uses Pedersen commitments with Bulletproofs-style range proofs (a proven, audit-ready cryptographic method); a full **zk-SNARK** upgrade is planned for production, not built yet.

Buyer Identity Binding & Attestation: A receivable can't go live on the seller's word alone. The buyer must confirm the debt through a verified digital identity (**tied to Bangladesh Bank's e-KYC framework**), not just a signature. This closes the loophole where someone could invent a fake buyer to validate a fraudulent invoice. The Olympiad prototype uses a clearly labeled mock version of this check.

Lender & Regulator Dashboards: Each side gets a live view suited to its role. Lenders see their own claim history as what's active, what's settled, what capacity checks they've run and real-time status on receivables they're involved in. Regulators see the full consortium picture: which invoices are flagged for review vs. clean, on-chain activity logs, and linked-entity risk alerts, all without seeing any lender's pricing or client details.

Validator Participation Bond (proposed): Lenders that want to run a validating node would stake collateral **(illustratively ~BDT 1,000,000)** as a commitment to honest participation. This mechanism is a design proposal, not part of the current MVP. Its **structure, custody, and enforcement** would need separate regulatory and legal approval before implementation.

Beneficial-Ownership Contagion Flag: Bangladesh's biggest financing frauds (Hallmark-Sonali, Crescent Group, the 2026 Premier Bank case) involved networks of separate companies secretly controlled by the same people, not one company reusing its ID. So instead of just matching business registration numbers, FinClaim flags claims linked through shared directors, signatories, or addresses. A flag triggers human review, not an automatic block.

Public Audit Anchor: Separately from the dashboards, the system periodically publishes a cryptographic fingerprint (Merkle root) of the ledger's history to a public blockchain. This lets outside auditors confirm the consortium's records haven't been secretly altered - without seeing any private data.

Redundant Off-Chain Evidence Storage: Encrypted invoices aren't kept in one shared location. Each institution stores its own encrypted copy, all linked by the same cryptographic fingerprint. No single storage provider can lose or withhold evidence during a dispute.

No Public Crypto Asset: FinClaim's "claim token" is just a registry entry - not a cryptocurrency, not something SMEs buy or trade, and not a way around Bangladesh's financial regulations.

Overview of the Competition

FinClaim has both adjacent and direct conceptual competitors. Contour/Corda-based LC projects digitize trade-document workflows and have already been used in Bangladesh, but they focus on letters of credit rather than a domestic receivable claim registry. Bangladesh Bank's CIB provides borrower-level credit information, not an asset-level "who has claimed this receivable?" state. MonetaGo is the strongest international benchmark because it directly targets duplicate financing and privacy-preserving verification. FinClaim therefore does not claim to invent invoice registries; its competition case rests on a Bangladesh-specific operating model and the additional mechanisms described above.

Overview of Risks

Adoption Risk: The registry creates value only when enough lenders participate; a small network can leave fraud paths outside the consortium.

Regulatory and Legal Risk: A ledger record does not automatically create legal priority or enforceability. Factoring, assignment, privacy and evidentiary rules must be aligned with Bangladesh law and regulator guidance.

Oracle and Collusion Risk: A forged or collusive buyer confirmation can still create a bad receivable. Blockchain cannot make false real-world data true.

Privacy Risk: Even if raw invoices stay off-chain, transaction timing and repeated entity activity can leak business metadata.

Technical Execution Risk: Bulletproofs-style range proofs, smart contracts and key management introduce implementation complexity that can create new vulnerabilities. A full zk-SNARK circuit, planned for production, adds further complexity on top of that.

False-Flag Risk: Linked-entity alerts may produce innocent associations; automatic blocking could unfairly freeze legitimate finance.

Mitigation Strategies

FinClaim uses a phased pilot, minimal on-chain data, independent security review, buyer attestation, role-based access, and manual review of contagion alerts. Full legal and production integration is deliberately separated from the olympiad prototype.

Regulator-led Pilot: Approach Bangladesh Bank's RFFO for a controlled pilot with 3-5 willing banks/NBFIs/fintech lenders before any national-scale rollout.

Phased Privacy: Prototype the claim-capacity logic first. Demonstrate a privacy-preserving YES/NO interface, then move to a fully audited zero-knowledge circuit before production.

Multi-source Verification: Require buyer confirmation through a credentialed DID, not a bare signature, and allow authorized external checks. Mark unavailable data sources clearly rather than pretending a mock oracle is a live government integration.

Dispute Governance: Use a frozen/disputed state, multi-party signatures and regulator audit access. No participant should be able to delete history.

Security Engineering: Use enterprise key management/HSMs in production, contract audits, penetration testing, role separation, incident response and recovery procedures.

Financials & Business Model

Our financial plan is built on a Public Infrastructure (Regulatory SaaS) model. In this setup, FinClaim operates as a central technology utility. We license the platform to Bangladesh Bank, and the commercial banks and large businesses that use the network pay for its services.

This model keeps our costs low while creating predictable, recurring revenue.

Revenue Streams

Verification Fees (Primary Income)

Every time a bank or lender checks the registry to see if an invoice is already claimed, they pay a flat query fee of BDT 100 to 300. This is a tiny cost for a bank to prevent a massive bad loan, but it scales up quickly for us as daily transaction volumes grow.

Penalty & Flag Removal Fees

If a business or lender is caught trying to double-finance an invoice, the system automatically flags them. To remove this red flag and restore their standing, the responsible party must pay a penalty fee equal to 0.02% of the flagged invoice's value.

Enterprise Subscriptions

Large corporate buyers (like major garment manufacturers) who upload thousands of approved invoices daily directly from their accounting software (ERP) pay a fixed, monthly subscription fee for seamless, high-volume access to the platform.

Cost Structure

System Development & Maintenance: Paying our core engineering and security team.

Hosting & Cloud Infrastructure: Running the central APIs and off-chain data storage.

Note on Cost Savings: Because we use blockchain and Zero-Knowledge Proofs, the heavy lifting of data processing happens on the banks' own computers (their "nodes"). This means our central server costs stay extremely low, even as the network grows to millions of invoices.

Scaling Strategy

Phase 1: Pilot & Proof (Months 1–12)
We launch with just 2-3 partner banks and a few enterprise buyers. Revenue is low, mostly coming from early enterprise subscriptions. The goal here is proving the tech works with zero false alarms.

Phase 2: Regulatory Mandate (Months 12–24)
We work with Bangladesh Bank to make the FinClaim registry check a mandatory step for all factoring loans. Once this happens, all banks must connect, and our Verification Fees skyrocket, making the platform profitable.

Phase 3: Regional Trade (Years 2–4)
We expand the system to handle cross-border export/import invoices. This brings in international banks and massive enterprise buyers, scaling all three of our revenue streams exponentially.

2-Year Financial Projections

Values listed in BDT Lakhs: **1 Lakh = 100,000 BDT.**

Category	Year 1 (1st Half)	Year 1 (2nd Half)	Year 2 (1st Half)	Year 2 (2nd Half)
Growth Metrics				
Invoices Verified (Volume)	1,000	5,000	20,000	75,000
Active Enterprise Subscribers	2	5	15	30
Revenue (BDT Lakhs)				
Verification Query Fees	2.0	10.0	40.0	150.0
Enterprise ERP Subscriptions	6.0	15.0	45.0	90.0
Penalty & Flag Removal Fees	0.5	1.0	2.0	5.0
Total Revenue	**8.5**	**26.0**	**87.0**	**245.0**
Operating Expenses (BDT Lakhs)				
Servers & Cloud Infrastructure	10.0	15.0	25.0	40.0

Core Development & Operations	25.0	35.0	50.0	70.0
Compliance & Legal Integration	15.0	10.0	25.0	40.0
Total Expenses	**50.0**	**60.0**	**100.0**	**150.0**
Net Profit / (Loss)	**(41.5)**	**(34.0)**	**(13.0)**	**+95.0**

Financial Highlights

Year 1 (The Pilot Phase)

We operate at a deliberate loss. Our main costs are building the core technology, setting up secure cloud infrastructure, and handling legal integrations with early pilot banks. Revenue is slow because we are only testing the system.

Year 2 - First Half (Scaling Up)

We are still operating at a loss, but the gap is closing. We spend heavily on development and compliance to ensure the system is ready for a national rollout.

The Break-Even Point (Month 21)

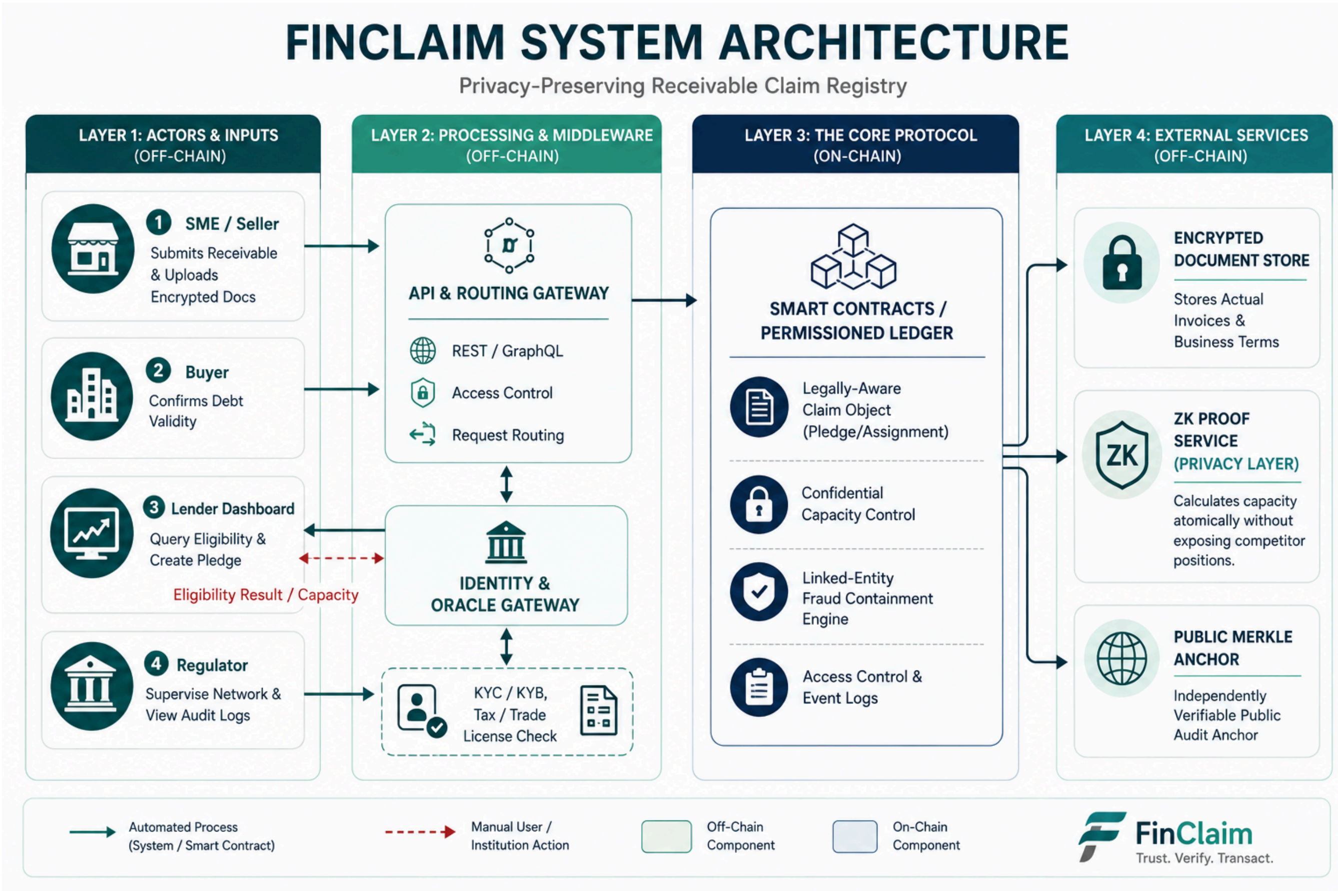
Halfway through the second half of Year 2, the Bangladesh Bank regulatory mandate kicks in. All banks are required to start checking invoices through FinClaim. The massive jump in verification volume (up to 75,000+) covers our operational costs completely.

Year 2 - Second Half (Profitable)

Because our blockchain structure shifts the heavy data processing onto the banks' nodes, our server costs don't spike when volume explodes. We finish the 24th month highly profitable, with a highly scalable and proven business model.

Business Model Canvas:

Prototype: Implementation



The roles of each module are as follows:

SME / Seller Portal: Submits receivable metadata and encrypted invoice evidence; sees verification and settlement status.

Buyer Attestation: The buyer confirms that the receivable exists and the payment obligation is recognized before activation.

Lender Interface: Banks/NBFIs/fintechs query eligibility and submit pledge or assignment claims without needing full visibility into competitors’ positions.

Backend / API Gateway: Handles authenticated requests, document workflow, integrations and non-consensus application logic.

Oracle Gateway: Connects authorized identity, BIN, tax/VAT or credit checks where legally/technically available. The olympiad prototype uses clearly labelled mock services.

Permissioned Ledger + Smart Contracts: Stores the canonical receivable state, signed claim events, status transitions and capacity rules.

Encrypted Off-chain Store: Keeps invoices, contracts, notices and sensitive commercial details encrypted; only fingerprints/commitments are referenced on-chain.

ZK / Confidentiality Service: Produces or verifies a proof that a proposed financing amount is within remaining capacity without disclosing unnecessary values.

Public Anchor: Publishes a periodic Merkle root or timestamp commitment so outside auditors can verify historical integrity without joining the private network.

Governance and Trust

- Only approved institutions can run participant nodes or submit financing claims; identities are tied to institutional certificates.
- Smart contracts enforce one shared lifecycle and atomic capacity updates for every participant.
- Bangladesh Bank anchors the consortium. It doesn't run it alone, and it isn't a commercial lender on the network. Freezing a claim, unfreezing it, or resolving a dispute needs sign-off from Bangladesh Bank plus a rotating panel of consortium members - not one institution acting unilaterally.
- No institution can silently delete or rewrite a settled claim. Corrections occur through new signed events that preserve the prior history.
- A public Merkle anchor adds external tamper evidence without exposing private invoice data.
- FinClaim uses no native coin and no public speculative token. Any institutional participation bond or penalty mechanism would require separate regulatory/legal design and is outside the MVP.
- Disputed claims can enter a FROZEN/DISPUTED state. Release, reversal or write-off requires authorized signatures and a visible audit reason.
- The ledger's security model assumes a bounded number of participating nodes can be faulty or colluding at any time (a standard permissioned-BFT assumption - e.g., tolerating f-of-3f+1), not that collusion is impossible. Public Merkle-root anchoring occurs on a fixed cadence (e.g., hourly); history within an unanchored window carries residual risk, which is disclosed here rather than left implicit.

Privacy and Security Risks

Privacy:

Is business and identity privacy addressed by the solution?

Yes, by data minimization rather than by publishing the invoice. Raw documents, prices and contracts remain encrypted off-chain. The ledger stores identifiers, hashes/commitments, authorization, state and timestamps. Institution-level access is permissioned.

Can a lender verify capacity without seeing competitors’ exact financing positions?

Yes, in the prototype itself, not just as a target. It uses Pedersen commitments with Bulletproofs-style range proofs to prove existing claims plus a new one staying under the receivable's value, returning TRUE/FALSE without revealing the numbers. A full **zk-SNARK** circuit is the production upgrade once this version has been through a pilot.

Crypto-security:

How are cryptographic keys handled?

Production deployment should use enterprise key management or hardware security modules, institutional certificate rotation and recovery procedures. Demo wallets are acceptable only for the prototype and must be labelled as such.

Access Control:

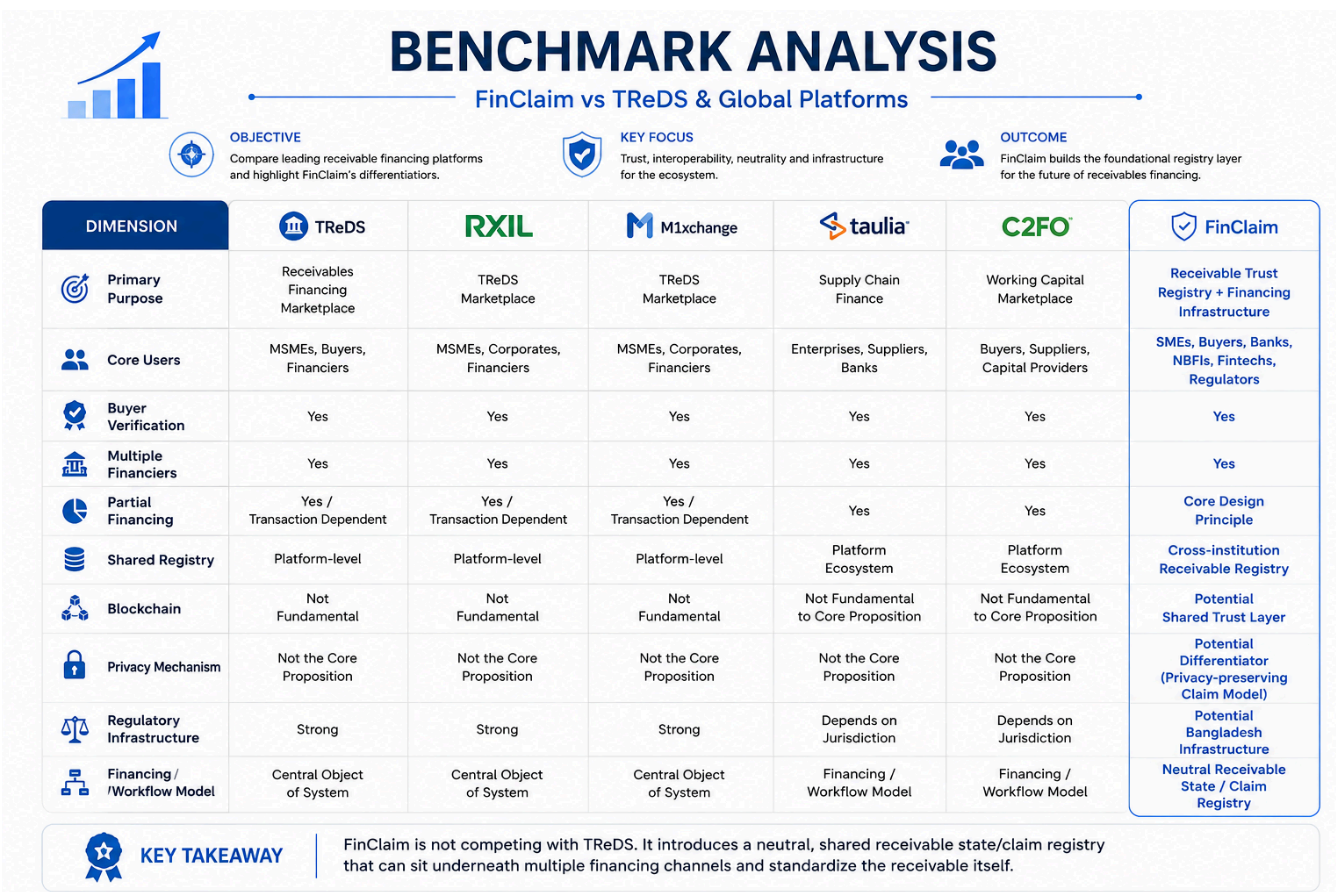
Who can see or change what?

SMEs can view their own receivables; buyers can attest to obligations involving them; lenders can query and claim according to role permissions; the supervisor can audit and freeze under defined governance. Independent auditors only need the public anchor to verify ledger integrity, not the private transaction contents.

Residual Security Risks:

Blockchain does not remove phishing, compromised keys, corrupt insiders, bad oracle data or legal disputes. FinClaim therefore combines cryptographic controls with institutional governance, audit, incident response and human review.

Benchmark Analysis



BENCHMARK ANALYSIS

FinClaim vs TReDS & Global Platforms

OBJECTIVE: Compare leading receivable financing platforms and highlight FinClaim's differentiators.

KEY FOCUS: Trust, interoperability, neutrality and infrastructure for the ecosystem.

OUTCOME: FinClaim builds the foundational registry layer for the future of receivables financing.

DIMENSION	TReDS	RXIL	M1xchange	taulia	C2FO	FinClaim
Primary Purpose	Receivables Financing Marketplace	TReDS Marketplace	TReDS Marketplace	Supply Chain Finance	Working Capital Marketplace	Receivable Trust Registry + Financing Infrastructure
Core Users	MSMEs, Buyers, Financiers	MSMEs, Corporates, Financiers	MSMEs, Corporates, Financiers	Enterprises, Suppliers, Banks	Buyers, Suppliers, Capital Providers	SMEs, Buyers, Banks, NBFIs, Fintechs, Regulators
Buyer Verification	Yes	Yes	Yes	Yes	Yes	Yes
Multiple Financiers	Yes	Yes	Yes	Yes	Yes	Yes
Partial Financing	Yes / Transaction Dependent	Yes / Transaction Dependent	Yes / Transaction Dependent	Yes	Yes	Core Design Principle
Shared Registry	Platform-level	Platform-level	Platform-level	Platform Ecosystem	Platform Ecosystem	Cross-institution Receivable Registry
Blockchain	Not Fundamental	Not Fundamental	Not Fundamental	Not Fundamental to Core Proposition	Not Fundamental to Core Proposition	Potential Shared Trust Layer
Privacy Mechanism	Not the Core Proposition	Not the Core Proposition	Not the Core Proposition	Not the Core Proposition	Not the Core Proposition	Potential Differentiator (Privacy-preserving Claim Model)
Regulatory Infrastructure	Strong	Strong	Strong	Depends on Jurisdiction	Depends on Jurisdiction	Potential Bangladesh Infrastructure
Financing / Workflow Model	Central Object of System	Central Object of System	Central Object of System	Financing / Workflow Model	Financing / Workflow Model	Neutral Receivable State / Claim Registry

KEY TAKEAWAY: FinClaim is not competing with TReDS. It introduces a neutral, shared receivable state/claim registry that can sit underneath multiple financing channels and standardize the receivable itself.

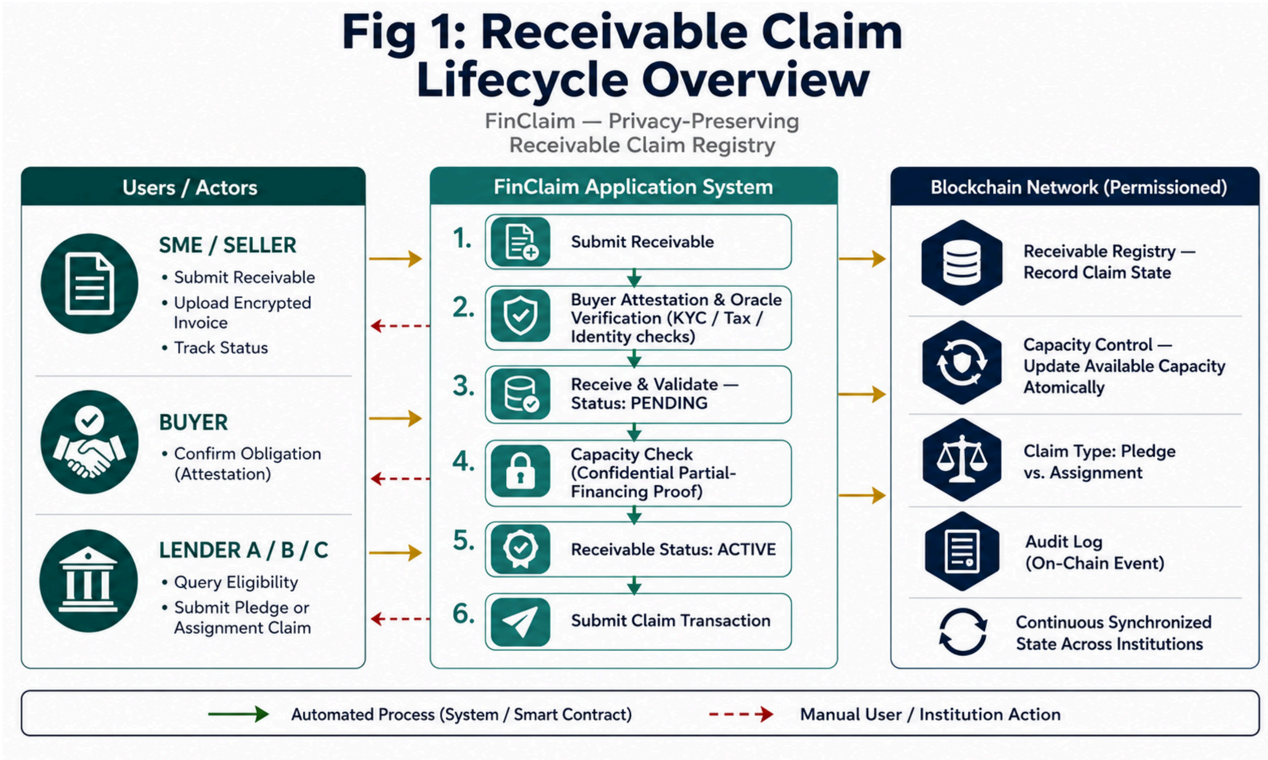
Prototype: Adherence to Decentralized Application Design

The FinClaim prototype is meticulously engineered to pass the core litmus test of decentralization: *If a centralized database or a single trusted operator could resolve the friction without introducing a single point of failure or conflict of interest, the inclusion of a blockchain would be superfluous.*

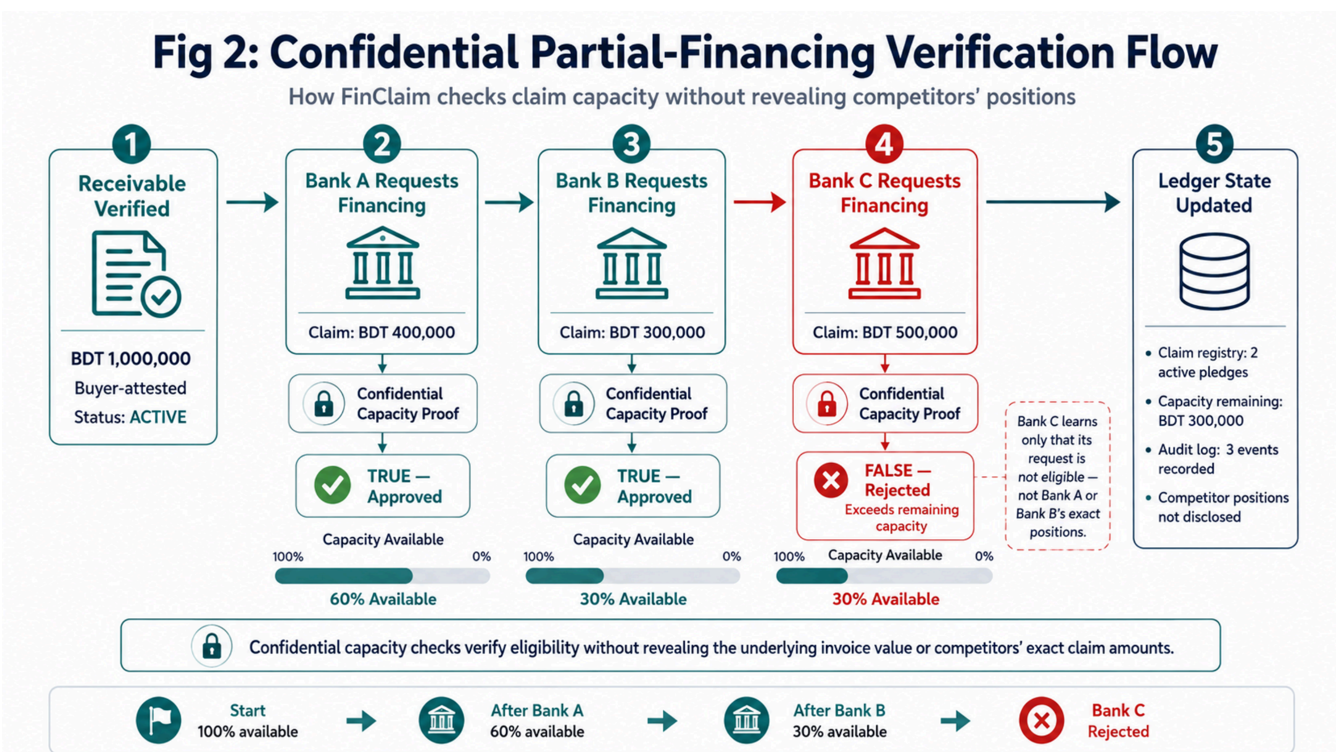
FinClaim avoids the pitfalls of legacy enterprise databases by adhering strictly to modern Decentralized Application (DApp) design principles. The architecture ensures that no single financial institution or regulatory body can monopolize data, manipulate financing states, or extract rent from the network.

The prototype's adherence to DApp architecture is demonstrated across the following protocol layers:

- **Neutral State Management (Consortium Consensus):** Unlike a traditional application where a central server dictates truth, FinClaim operates as an EVM-compatible permissioned consortium. Competing lenders (Banks, NBFIs) and the regulator act as validating nodes. State transitions - such as moving a claim from *Verified* to *Pledged*—are executed through consensus protocols, ensuring that the ledger is a shared, neutral checkpoint rather than a proprietary silo.
- **Deterministic Smart Contract Execution:** The core business logic is entirely abstracted from backend APIs and embedded directly into smart contracts. The "Legally-Aware Claim Object" programmatically enforces the distinction between a pledge and an assignment. Furthermore, the partial-financing logic runs deterministically on-chain; the smart contract atomically calculates available claim capacity and prevents over-financing without requiring manual intervention or central administrative bias.
- **Zero-Knowledge Privacy-by-Design:** A fundamental failure of early DApps was the broadcasting of sensitive commercial data to all network participants. FinClaim adheres to privacy-preserving architectural standards by utilizing Zero-Knowledge (ZK) Proofs (e.g., Pedersen commitments). A lender queries the smart contract to verify if an invoice has available capacity, and the protocol mathematically proves an eligibility result without exposing the exact financial positions, client identities, or lending volumes of competitor banks.
- **Decentralized & Off-Chain Storage Orchestration:** True to Web3 scaling principles, the FinClaim blockchain does not bloat its ledger with raw data. All heavy, commercially sensitive documents (such as actual invoices, trade contracts, and PDF attachments) are encrypted and stored in secure off-chain object storage. Only the immutable cryptographic hashes of these documents are anchored on the ledger, guaranteeing document integrity and tamper-evidence without violating data sovereignty.
- **Trustless External Data via Oracle Gateways:** Because a blockchain cannot natively verify real-world events, the DApp architecture utilizes secure Oracle gateways. Before a receivable can be converted into an active claim object, external identity verification (e-KYC/KYB), trade license validation, and explicit Buyer Attestations are securely bridged onto the ledger via Oracles. This ensures the on-chain digital claim is inextricably linked to real-world legal obligations.
- **Public Merkle Anchoring for Verifiability:** To eliminate the risk of a consortium majority colluding to secretly alter historical records, FinClaim incorporates an independent audit layer. The root hashes of the permissioned consortium blocks are periodically anchored to a public, permissionless blockchain. This hybrid design provides the high throughput and privacy of a permissioned DApp while inheriting the ultimate immutable security of a public ledger.



The flow starts with a seller submitting a receivable, but FinClaim does not mint a financeable claim immediately. The buyer must first confirm the obligation, and available external checks are requested. Once verified, the receivable becomes ACTIVE. A lender can then request a pledge or assignment. If the rules pass, the transaction is written to the shared ledger and the available claim capacity is updated atomically. This means every participating institution sees the same state even though sensitive commercial details remain private.



The partial-financing example is the prototype's central demo because it makes the problem visible in seconds. A BDT 1,000,000 receivable can safely support several financing allocations as long as their total remains within the verified claim capacity. The contract accepts Bank A's BDT 400,000 and Bank B's BDT 300,000, but rejects Bank C's BDT 500,000 request. In the target privacy design, Bank C does not need to learn the exact earlier allocations; it only learns that its request is not eligible.

When the buyer settles the receivable, financing allocations are marked settled/released and the receivable closes. If fraud is confirmed instead, FinClaim creates a dispute event and activates linked-entity review. Other outstanding claims connected to the same verified business identifier are surfaced to authorized reviewers, helping contain a pattern that may span multiple invoices or related entities.

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